# APPENDIX 1

3BLEVY

## SOLVED EXAMPLE UNIT-1

1. Find the dimensions of a and b in the formula $[P + \frac{a}{V^2}][V - b] = RT$ where P is pressure and V is the volume of the gas

**Solution:**

By the principle of homogeneity, $a/V^2$ is of the dimensions of pressure and b is of the dimensions of volume.

$$[a] = [\text{pressure}]\ [V^2] = [ML^{-1}T^{-2}]\ [L^6]$$
$$= [ML^5T^{-2}]$$
$$[b] = [V] = [L^3]$$

2. Show that $(P^{-5/6}\rho^{1/2}E^{1/3})$ is of the dimension of time. Here P is the pressure, $\rho$ is the density and E is the energy of a bubble)

**Solution:**

Dimension of Pressure $= [ML^{-1}T^{-2}]$

Dimension of density $= [ML^{-3}]$

Dimension of Energy $= [ML^2T^{-2}]$

By substituting in the given equation,

$$= [ML^{-1}T^{-2}]^{-5/6}[ML^{-3}]^{1/2}[ML^2T^{-2}]^{1/3}$$
$$= M^{-5/6+1/2+1/3}L^{5/6-3/2+2/3}T^{5/3-2/3}$$
$$= M^0L^0T^1 = [T]$$

3. Find the dimensions of mass in terms of Energy, length and time

**Solution:**

Let the dimensions of Energy, Length and Time be $[E],[L],[T]$ respectively.

We know that Force = mass x acceleration

$$\text{Mass} = \frac{Force}{acceleration}$$
$$= \frac{Workdone\ (or)\ Energy}{acceleration \times displacement}$$
$$[m] = \frac{Energy}{[acceleration][displacement]}$$
$$= \frac{[E]}{[LT^{-2}][L]} = \frac{[E]}{L^2T^{-2}} = [EL^{-2}T^2]$$

4. A physical quantity Q is found to depend on quantities x,y,z obeying



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